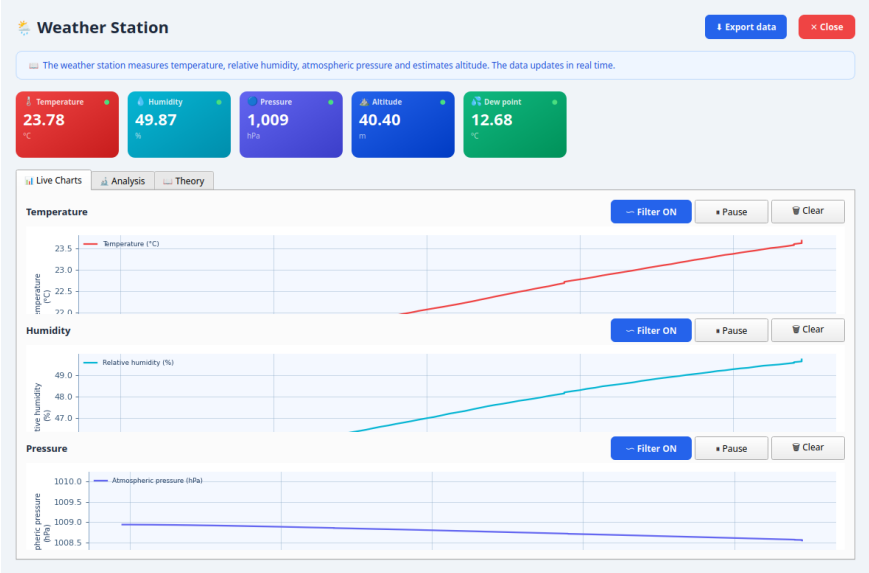
Weather Station

A forecast rests on the pressure trend and on context, not on a threshold

Level: Explorer to Scientist — ages 11–18 (Romanian school years V–XII) • **Time in class:** 20 min, one 50-minute lesson or a two-week project • **Sensors:** BME280 (temperature, humidity, pressure, altitude)



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Explorer to Scientist — ages 11–18 (Romanian school years V–XII); in the app the activity is tagged **11+**, because its theory tab shows formulas with a natural logarithm (at Junior level the dew-point formula is hidden). The same app serves both levels, but the task differs: at Explorer, students read the *direction* of the trend ("rising / falling / steady"); at Scientist, they compute the trend in hPa per hour from the exported file and score their forecasts. The most direct curriculum entry points are **year V** (geography, the Earth's atmosphere) and **year IX** (general physical geography: the synoptic chart and forecasting). The Magnus formula, with its logarithm, is to be read, not calculated, before year X.

Assumed knowledge

Reading a chart over time and calculating a difference. For the full route, in addition: a rate (change divided by time) and the arithmetic mean. For the project: working in a spreadsheet. For support: only "rising / falling / steady".

Time in class

Demonstration: 20 min. Full lesson: 50 min. Project: two school weeks — two 50-minute lessons, one 15-minute check and eight days of 10 minutes of routine, that is 195 minutes in total (3 h 15 min).

Teacher preparation

15 min — choosing and labelling the fixed spot for the station, one trial day with the exported file open at the end, finding the school's true altitude, the file-naming rule and the shared folder, printing the sheets and the trend chart for students on the support route.

Sensor settling time	**10 minutes after any move** or after power-on — temperature and humidity are the slowest. At the station's fixed spot, where the box no longer moves, there is no settling time left, and that is precisely the argument for a fixed station.
Sensors and boxes	BME280. One box per class is enough for the project; for comparison investigations (indoor/outdoor, microclimate) two are useful, but **the same box must be used for the whole of one series**.

Three routes

Short	20 min: projected demonstration — the five cards, breathing from 30 cm, one trip outside and one storey climbed, then the export.
Full	50 min: a single investigation from the lesson-flow section, run to completion — prediction, measurement, export, first analysis.
Project	Two weeks: fixed station, fixed timetable, one folder and one cumulative project file, a daily forecast written down and compared with the official one.

1. Where this fits in the curriculum

The app turns a MagicBox into a working automatic weather station. It shows five live values — temperature, relative humidity, atmospheric pressure, estimated altitude and dew point — above three scrolling charts, plus an Analysis tab that translates each reading into words. Everything exports to a dated file, which is exactly what makes genuine day-to-day meteorology possible. The activity fits:

- **Geography** (*Geografie*) — at **year V**, the Earth's atmosphere: air temperature, atmospheric pressure, humidity, weather and climate. At **year IX**, general physical geography covers **the synoptic chart and forecasting**, and the activity produces, with classroom instruments, exactly the quantities shown on that chart. The school playground's microclimate, compared with the county forecast, is the most direct application of the idea of geographic scale.
- **Physics** (*Fizică*) — at **year VI**, temperature and thermal equilibrium; at **year VII**, **atmospheric pressure**, explicit syllabus material; at **year X**, molecular physics and thermodynamics give saturation vapour pressure and condensation, the background to the dew point.
- **Mathematics** (*Matematică*) — the arithmetic mean and the range of a data series at **years V–VI**; plotting points on a coordinate system at **year VII**; and at **year VIII**, the gradient of a line and the first-degree function — **the pressure trend is a gradient**, and this is the cleanest meeting of the mathematics syllabus with a real phenomenon in the whole collection.
- **Computer science and ICT** (*Informatică și TIC*) — **years V–XII**, processing data in a spreadsheet: computed columns, an XY chart with time on the horizontal axis, daily means, overlaying two days. The exported file opens directly.
- **Cross-curricular** — the difference between a *value* and its *rate of change*, correctly stating an instrument's exposure, the honesty of scoring a forecast even after it fails, and the habit of recording conditions alongside a number.

The core skill is **reading a trend rather than a value**. A forecaster does not ask "what is the pressure?" but "which way is it going, and how fast?" That shift — from a number to its rate of change — is the intellectual heart of the activity, and it is exactly what a single reading can never teach.

2. Time plan

Route	Time	What happens
Short	**20 min**	Projected demonstration. The five cards read out loud (4), gentle breathing *near* the box from 30 cm, with the jump in humidity and dew point (4), one trip outside or beside an open window (5), climbing one storey to watch the pressure fall (4), the export and opening the file (3).
Full	**50 min**	A single investigation from the next section, run to completion: the written prediction and discussing it (8), setting up and the 10-minute settling time, used for theory (12), the measurement (15), the export and the first analysis (15). Investigation 2 (indoor/outdoor) and investigation 3 (the height of the building) work best.
Project, day 1	**50 min**	The five quantities, the idea of a trend, the daily routine practised by everyone (reading, exporting, renaming), the first forecast written as a class.
Project, days 2–9	8×10 min	The duty pair: checks the stream, copies the five values onto the wall chart, presses **Export data**, renames the file with the date, writes one forecast sentence and records yesterday's actual weather.
Project, day 5	15 min	Mid-project check: is the routine holding? Are the filenames consistent? Score the forecasts so far.
Project, day 10	**50 min**	All the files loaded into one spreadsheet, the fortnight of pressure as one continuous line, the rain days marked on it, the forecast score, and the comparison with the official forecast.

The project means **two 50-minute lessons** (the first and the last day), one **15-minute** check in the middle and **eight days of 10 minutes** of routine — that is $50 + 50 + 15 + 8 \times 10 = 195$ minutes, **three and a quarter hours** spread over two school weeks. It is the only activity in the collection that asks for patience instead of one concentrated lesson, and that is exactly what the app was built for.

The daily routine has to cost 10 minutes, not 30

Appoint a duty pair, with a weekly changeover. Their whole task, at a fixed time, is: check the box is streaming, copy the five values onto the wall chart, press **Export data**, rename the file with the date, write one forecast sentence and record how yesterday's weather actually turned out. Ten minutes. A project that costs half a lesson every day will be quietly abandoned in week two — and an abandoned project has no data to analyse, so it has no lesson either.

3. Materials

What you need	Per class	Notes
MagicBox with BME280	1	Provides all five displayed quantities. For a two-week series, use **the same box from start to finish**; two units can differ by a few tenths of a degree and by 1–2 hPa.
Laptop with the app	1	For the fixed station it stays put during reading; for the travelling investigations it goes with the group.
A safe fixed spot for the station	1	Shaded, out of direct sunlight, away from a radiator, a door and a laptop's warm air, at a steady height. A high shelf or a window recess with the blind partly down works well. **Do not move it again until the project ends.**
Extension lead or power bank	1	Needed for a station running for hours or overnight. Check the cable and route it safely.
Ordinary wall thermometer	1	A daily cross-check. A mismatch is a lesson, not a fault.
Wall chart, clipboard and labels	1 set	The paper record students look at daily; the label marks the station's exact position, so it can be put back identically if someone moves it.
Tape measure	1	For a storey's height and the station's height above the floor.
Printed local forecast, daily	1	From the national weather service. The class forecast is compared against it *and* against what actually happened.
The school's true altitude	1	From a map. Without it you cannot explain why the app's pressure card says what it says.
Shared folder or USB stick	1	All the dated files live in one place, together with **the project's cumulative file**.

Preparation: choose and label the station's position before the first lesson and do not move it for the duration of the project — the whole value of the data set rests on this. Let the box run for a full day beforehand and look at the exported file, so you know what normal looks like in your room. Find out the school's altitude above sea level; without it you will not be able to explain to the class what the pressure card shows. Agree the file-naming rule and set up the shared folder before day one. For students on the support route, print **the trend chart already drawn** — see the adaptations box.

4. Scientific background for the teacher

The forecast rests on the trend and on context, not on the 1013 hPa threshold

Atmospheric pressure at a point is the weight per unit area of the column of air above it. The standard value at sea level is

$$P_0 = 1013.25 \text{ hPa},$$

and at ground level values between about 980 hPa in a deep depression and about 1035 hPa

in a strong anticyclone are typically found. The absolute value, however, matters far less than the **tendency** — the change over the previous three hours:

$$\text{tendency} = \frac{P(t) - P(t - 3\,\text{h})}{3\ \text{h}} \qquad [\text{hPa/h}].$$

Real tendencies are small: on an ordinary day pressure changes by 0.1–0.3 hPa per hour, so almost nothing shows in ten or fifteen minutes — the tendency is judged over **three hours**. Meteorologists describe it with a few fixed words — **steady**, **rising**, **falling**, **rapid** — with indicative thresholds like these: under about 1 hPa in 3 hours, practically steady; 1.5 to 3.5 hPa in 3 hours, a clear rise or fall; above about 3.5 hPa in 3 hours (more than 1 hPa per hour), a rapid change. A sustained fall usually means an approaching front, with cloud, wind and rain; a slow rise spread over many hours means a building anticyclone, settled weather, often clear.

The daily pressure cycle. Even with no change in the weather, pressure rises and falls twice a day, by up to about one hPa between maximum and minimum (maxima around 10:00 and 22:00, minima around 04:00 and 16:00). A fall of half a hPa between 10:00 and 16:00 may therefore be only this atmospheric "tide"; that is why the tendency is compared with the previous days at the same times. **The slope of the chart is the forecast**, and that is why the app's pressure chart is more useful than the pressure card.

Why the 1013 hPa threshold cannot work. The app shows the pressure *at the station*, not reduced to sea level. Because pressure falls by about 0.12 hPa per metre of altitude, a school at 300 m reads $300 \times 0.12 = 36$ hPa less than a school at sea level in exactly the same weather situation: $1013 - 36 = 977$ hPa. Such a school would, by the threshold, be in a storm every day of the year, including in the middle of a high-pressure system. Conversely, a school at sea level can show 1010 hPa under a completely clear sky. The threshold does not tell weather apart from relief. Weather maps compare stations only after each station's pressure has been *reduced to sea level*, adding about 0.12 hPa for every metre of altitude; the app does this reduction too, with the site altitude from **Settings**.

Context. The tendency alone is not a forecast either. A meteorologist also has: the direction the wind is coming from, cloud type, the season, the synoptic chart of a whole region, and numerical models. The class has only the tendency — and that is exactly why its score will be good, but not very good. This is an honest conclusion, not a failure.

4.1 Relative humidity, dew point, and why they move in opposite directions

Warm air can hold more water vapour than cold air. Relative humidity is the ratio of the actual vapour pressure e to the saturation vapour pressure $e_s(T)$ at the current temperature:

$$\text{RH} = \frac{e}{e_s(T)} \times 100\%.$$

Saturation vapour pressure roughly doubles for every 10 °C rise, which has a consequence students find counter-intuitive: **heating air lowers its relative humidity without removing a single drop of water from it**. This is exactly why classrooms are dry in winter: outdoor air at 2 °C and 90% RH, heated to 21 °C, arrives at about 26% RH, despite holding exactly the same water.

The dew point T_d is the temperature to which air must be cooled, at constant pressure and water content, for condensation to begin. The app computes it with the Magnus approximation, with the same constants in every activity:

$$\gamma = \frac{aT}{b+T} + \ln\left(\frac{\text{RH}}{100}\right), \qquad T_d = \frac{b\,\gamma}{a-\gamma}, \qquad a = 17.625, \quad b = 243.04\,^\circ\text{C}.$$

Worked example for an ordinary classroom, $T = 22.0$ °C and RH = 45%: $\gamma = 17.625 \times 22/(243.04 + 22) + \ln 0.45 = 1.463 - 0.799 = 0.664$, so $T_d = 243.04 \times 0.664/(17.625 - 0.664) = 9.5$ °C. The formula

needs the natural logarithm, which is taught in year X; at Junior level the app does not show it, and in lower-secondary school students read the dew point off the card and interpret it, without calculating it. The *deficit* $T - T_d = 12.5\,°\text{C}$ shows comfortably dry air; as the deficit approaches zero, fog, dew and condensation appear.

Unlike relative humidity, the dew point is an *absolute* measure of how much water is in the air. It barely changes when you carry the box from a cold corridor into a warm room, while relative humidity changes dramatically. This contrast is the most valuable demonstration in the whole app.

4.2 The altitude formula, and why the absolute number is wrong

Pressure falls with height because there is less air left above. The app uses the standard-atmosphere form:

$$h = 44330 \left[1 - \left(\frac{P}{P_0} \right)^{0.1903} \right] \text{ m},$$

which near sea level is very nearly linear, at about 0.12 **hPa per metre** — so a storey of 3.3 m gives $3.3 \times 0.12 = 0.40$ hPa, which the sensor resolves comfortably.

Two cautions are worth passing on to students. First, the formula uses the sea-level pressure from **Settings**, 1013.25 hPa by default, but the real sea-level pressure today may be 10–30 hPa away from that, so the *absolute* altitude reported can be ±100–250 m wrong, while the *difference* between two floors stays correct; with the day's QNH entered in **Settings**, the error drops to a few metres. Second, the **Altitude** card moves on its own, overnight, without anyone touching the box — a change of 15 m means $15 \times 0.12 = 1.8$ hPa of pressure, an entirely ordinary weather change. An uncalibrated barometric altimeter measures weather and height in equal measure; telling the two apart is precisely the job.

4.3 Why an instrument's exposure matters

Meteorological services keep thermometers in a ventilated white screen, 1.25–2 m above short grass and well away from buildings, for a very practical reason. A sensor in direct sunlight measures its own warmed casing; one near a radiator or a laptop's vent measures that object, not the air. When the class station disagrees with the official forecast, the first question is never "is the sensor broken?" but **"how is it exposed?"**

5. Lesson flow

The Weather Station is not a step-by-step wizard. Unlike most activities in the collection, there is no green button that advances it, no **Back**, no **Finish**, no observation log — it is a window that runs for as long as you leave it open. So the pace of the lesson is set by you, not by the app.

Technical note — what is on screen, and what buttons actually exist

How to start it. From the **Activities** page, with the **Start** button on the **Weather Station** card. The app requires a connected box; if none is connected, it refuses with a message asking you to connect a box before launching the activity. If more than one box is connected, a picker appears first, with one option for each box, under the name given to it in **Settings** (for example "Group 1"). With several suitable boxes, the **Change box:** list in the window header moves the station to another box without closing the window.
The five cards, in the order they appear on screen: **Temperature** (°C), **Humidity** (relative humidity, %), **Pressure** (hPa, to one decimal place; two at Scientist level), **Altitude** (calculated altitude, m, to one decimal place) and **Dew point** (°C).

The three tabs. Live Charts — three charts stacked one above the other, for temperature, humidity and pressure, on a shared time axis and with a single row of buttons, in the right-hand corner of the tabs: **Smooth: on / Smooth: off**, **Pause** and **Clear**. The charts show **the whole session**, from the moment the window opened, up to the last three hours at most — exactly the span over which the tendency is judged. The pressure chart always spans at least 1 hPa vertically, so that noise does not look like a change in the weather. **Analysis** — three rows that translate temperature, humidity and pressure into words, plus the barometric altitude formula. **Theory** — a page to read, with the dew-point formula (hidden at Junior level).
The button that matters: Export data, at the top right, beside **Close**, which saves the session's measurements as `.xlsx` (the default) or `.csv`. If you want the *picture* of a chart, press **Pause** to freeze it and take a screenshot — or, better, redraw the chart from the exported file.
How long to wait. Ten minutes after any move or power-on: temperature and humidity are the slow channels. At the fixed station, the problem disappears on its own, which is exactly the argument for a fixed station.

5.1 The short route: the 20-minute demonstration

Project the screen. Read the five cards out loud and ask the class to predict, *beforehand*, what will happen to each. Then three moves, in this order:

1. **Breathing from 30 cm (4 min).** Blow gently *near* the box, not directly onto it. **Humidity** and **Dew point** jump immediately; **Temperature** moves less. Ask why they jumped together. Then wait for them to return: it takes minutes, and that is exactly what needs to be seen.
2. **Outside, or beside an open window (5 min).** In winter, temperature falls and relative humidity rises, but **the dew point barely moves**. This is the central demonstration: the air has not gained water, it has cooled.
3. **One storey up (4 min). Pressure** falls by about 0.4 hPa, and **Altitude** climbs by 3–4 m. Show that the *difference* is good even though the altitude's absolute value does not match the map.

Close with the export: press **Export data**, save the file and open it in front of the class. A file opened once in front of everyone makes the two-week project credible.

5.2 The full route: five ready-made investigations

Each one fits into a 50-minute lesson, with a written prediction at the start and an export at the end. Choose *one*.

1. Pressure tendency as a forecast. *Can tomorrow's weather be forecast from the way today's pressure is moving?* The station stays in place all day; readings are taken at 09:00, 12:00 and 15:00. The duty pair computes the tendency over 3 and over 6 hours — and compares it with the previous day at the same times, so as not to mistake the daily cycle for a change in the weather — and writes a one-sentence forecast — *rain / cloud / no change / clearing* — plus a confidence out of 5. Over ten days a class typically scores 60–75%: better than chance, well short of a meteorologist, and an excellent basis for a discussion of what else forecasters use.

2. Indoors versus outdoors: the dew point stays put. *What actually changes when air enters the building?* Record T, RH and T_d outside, in the shade, for five minutes; go inside, wait ten minutes, and record again; then the other way round. Expected result, in winter: temperature rises a lot, relative humidity falls a lot, *the dew point moves by only a degree or two.* This is the clearest demonstration of what relative humidity actually means.

3. How tall is the school. *How tall is our building, measured with air?* Pressure at the ground floor for one minute, then on every floor, with a full minute of waiting on every landing, then **the return to the ground floor**. One storey checked with a tape measure. You get about 0.4 hPa per storey and jumps of 3–4 m on the **Altitude** card; the absolute altitude may be tens or hundreds of metres off, and the differences will be good — that is the point. The difference between the two ground-floor readings, usually 0.1–0.3 hPa, is a *measured* estimate of the uncertainty. *For the full version, with a tape measure and a calculation, use the activity* Pressure and Altitude.

4. The ventilation experiment. *What does opening a window actually do, and how long does it last?* Ten minutes of baseline with the class seated, five minutes with all the windows open, then twenty minutes with them closed. The box does not move at all, and the whole session is exported as one file. Temperature falls by 1–4 °C within a few minutes and recovers in 15–30; humidity and dew point move towards the outdoor values and recover more slowly. In winter the dew point falls and stays low — which is why a frequently ventilated room feels dry.

5. Our microclimate against the national forecast. *Does the official forecast describe our playground?* Every day, at the same time, the outdoor readings (box carried out and left for ten minutes) alongside the forecast temperature for your town, for two weeks. A consistent bias of 1–3 °C usually appears, almost always on the warm side, because a school playground is built up and rarely as well exposed as an official weather screen. The fact that the bias is *consistent* is the argument that the station is not wrong: it is measuring a different place. That is the definition of a microclimate.

5.3 The two-week project: the station, the timetable, the file

Three rules make the difference between a data set and a heap of numbers.

The station does not move. Not to make room for a display, not for an event. If it must move anyway, record the date and treat the series from that point on as a new, separate series. The position is marked with a label, so it can be put back identically.

The timetable is fixed. 09:00 and 15:00 every day. Indoor temperature follows the heating schedule and the number of people in the room, both of which depend strongly on the time of day; a reading at different times cannot be compared with anything. Every reading also records **the state of the room**: windows open or closed, heating on or off, how many people, blinds up or down. One line. *Most of the unexplained variation lives in this line.*

One folder and one cumulative file. All the daily exports go into a single folder — not one folder per week — and in the same folder sits **one project file**, a spreadsheet in which the duty pair copies, every day, one row: the date, the time, the five values, the tendency, the written forecast and yesterday's actual weather. The analysis on day ten is done on that file. Without it, ten daily files are ten files, not a data set.

Technical note — right at the step where the class opens the Analysis tab

The **Analysis** tab has three rows. The first two translate the reading into words, with fixed thresholds: temperature is **Below zero (freezing)** below 0 °C, **Very cold** below 10, **Cold** below 18, **Comfortable** below 24, **Warm** below 30 and **Very hot / heat wave** above; humidity is **Very dry (health risk)** below 20%, **Dry** below 40, **Optimal** below 60, **Humid** below 80 and **Very humid (mould risk)** above. These are reasonable comfort conventions, as far as they go.

The third row, the pressure row, works like a meteorologist. It no longer compares the station value with the fixed 1013 hPa threshold. It shows the **pressure reduced to sea level**, computed with the site altitude from **Settings** (the **Altitude Calibration** section), and the **tendency over the last three hours**, in the words meteorologists use: **steady** (under 1 hPa in three hours), **rising** or **falling**, and above 3.5 hPa **rising, rapid** or **falling,**

rapid. Until the station has three hours of data, the row reads "tendency: needs 3 hours of data (so far … min)" — which is why a window opened for just one lesson does not give a tendency, and the fixed-station project does.
What you need to set. If the site altitude in **Settings** is not the school's, the sea-level pressure comes out wrong by about 0.12 hPa for every metre of error. Check it once, at the start of the project, against the altitude on the map; then compare the class's sea-level pressure with the one in the national forecast — agreement within two or three hPa means a well-adjusted station.
What is left to discuss. The tendency on screen is correct, but understanding it does not come by itself. Show the class the station value, then the school's altitude, then work out the sea-level pressure together and compare it with the one on screen. Mention the daily cycle: an afternoon "fall" of half a hPa may be just the atmosphere's tide. And say plainly why it is the trend that matters, not the threshold — **"it matters where it is going, not where it is"** is worth writing on the board and leaving there for the whole project.

Technical note — the export, file naming and the automatic log

The button. Export data, at the top right. It opens a save dialogue, with a suggested name already built from the box's name (the one given in **Settings** or, if it has none, its identifier) and the date — for example `BOX_3859_2026-09-14.xlsx` — and two formats: **Excel workbook (*.xlsx)**, the default, and **CSV (*.csv)**. A CSV saved while the app is in Romanian, French or German uses ";" and a decimal comma, so it opens correctly in an Excel set to one of those languages; in English it uses commas and a decimal point. On success, the message **Export complete** appears; if nothing was recorded in the session, **No data** appears instead.
Two readings a day. Complete the name by hand: `BOX_3859_2026-09-14_am.xlsx` and `BOX_3859_2026-09-14_pm.xlsx`. Keep the date in `YYYY-MM-DD` form, so that sorting by name sorts by time.
The automatic log. The app writes, on its own, into the `logs` folder, a file of the raw stream as data arrives — a genuine safety net when a group forgets to export. Those files are **no longer deleted automatically**: automatic deletion is off by default and can only be switched on in **Settings**, under **Keep logs (0 = always)**. Check that the setting stays at 0 for the whole project. Even so, **what matters is exported and filed in the shared folder**, on the day itself: the log is the safety net, not the project archive.
The Data & Export page of the application, visible at Scientist and Teacher level, lists every recording and can export them, including every box at once, with **Export all boxes (Excel)**.
A note on the dew point, for anyone also running the *Physics of the Atmosphere* activity: the **Dew point** card here and the one in *Physics of the Atmosphere* now use the same Magnus constants (17.625 and 243.04 °C), so they give the same number for the same readings.

5.4 Weekends, holidays and handovers between classes

Weekend gaps are informative, not damaging: the Monday-morning reading, after two days with no heating and no people, is one of the most interesting points in the whole series and should be marked on the chart, not smoothed away. If a box can safely stay switched on over a weekend in an agreed spot — discuss this with the building manager first — do it: a continuous 72-hour pressure trace through a passing front cannot be obtained any other way. Over a holiday, stop the series cleanly, export it, and restart it as a new block, rather than pretending it is continuous.

For handovers between class groups, keep a laminated **station log** beside the box: the position, the height, the reading times, the naming rule and one line per day signed by the duty pair. Any group should be able to take over from that sheet alone.

The investigation, in one table

The question. Can tomorrow's weather be forecast from today's pressure trend, and how often do we get it right?
The hypothesis. Written on the sheet **before** the first day of measurement, in the form: *the pressure will* _____ *before it rains*, plus a number for how many forecasts out of ten we think we will get right. Not changed afterwards.
What I change. Nothing. This is an *observational* investigation: the only thing that changes is the weather, and we simply follow it. Worth saying explicitly to the class, because it is a kind of research just as legitimate as a controlled experiment.
What I measure. Pressure, in hPa, at fixed times; the derived tendency, in hPa per hour; and the actual weather the next day, recorded *as it was*, not as we predicted.
What I keep the same. The same box, **the same marked spot**, **the same times** every day, the same file-naming rule, and the same person who decides what "it rained" means. Define this criterion on day one and write it on the board: a class that decides halfway through the project what "rain" means has ruined its own score.
How many repeats. Ten forecasts is the minimum that says anything. Every day is a repeat; no day is discarded, however inconvenient it is for the story.
What makes a result valid — and what makes it an artefact. Compare the class's score with **the score of a lazy forecast**: "tomorrow will be like today". In a temperate climate, the lazy forecast gets it right on its own about 60% of days. **If the pressure tendency does not beat the lazy forecast, the correct conclusion is that your method added nothing — not that pressure has nothing to do with weather.** These are two different statements, and the distinction between them is exactly what is being taught here. Then also compare against the national weather service's official forecast for the same days: it has information the class does not, and the score gap measures exactly that information. A poor score obtained honestly, with the criterion fixed from the start, is worth more than a good score obtained by moving the criterion.

6. Analysis after the activity

6.1 What the data should look like

Quantity	Typical range	Notes
Indoor temperature	19–24 °C	Rises by 1–3 °C over an occupied lesson.
Outdoor temperature	−5 to +35 °C	Season dependent; shade only.
Relative humidity, indoor / outdoor	30–60% / 50–95%	Below 30% indoors in winter; near 100% outdoors in fog or rain.
Indoor dew point	5–14 °C	Changes far less than RH between two rooms. This is exactly the point.
Station pressure	960–1030 hPa	Depends *both* on the school's altitude *and* on the weather situation. At sea level, below 990 hPa means a depression, and 980 hPa a deep depression.
Tendency, settled weather	0.1–0.3 hPa/h	Practically flat; on screen, **steady**.
Daily cycle	up to about 1 hPa between maximum and minimum	Maxima around 10:00 and 22:00, minima around 04:00 and 16:00; not a change in the weather.
Tendency, an approaching front	−0.5 to −2 hPa/h (1.5–6 hPa in 3 h)	**Falling** or **rapid**; cloud and rain likely within a few hours.
Pressure change over one storey	0.35–0.45 hPa	About 0.12 hPa per metre.
The **Altitude** card's overnight jump	up to 15 m, box undisturbed	That is 1.8 hPa of weather, not 15 m of building.
Temperature drop on ventilation	1–4 °C in 5 min	Recovers in 15–30 min.
Forecast score over ten days	60–75%	Compare with the lazy forecast, not with zero.

6.2 How to compute the tendency, step by step

1. From the project's cumulative file, take the pressure at 09:00 and the pressure at 12:00 for the same day.
2. Tendency over three hours = (pressure at 12:00 − pressure at 09:00) / 3, in hPa per hour. Worked example: 1006.2 − 1008.4 = −2.2 hPa in three hours, so −2.2/3 = **−0.73 hPa/h** — a clear fall, several times an ordinary day's tendency, so cloud and rain are the forecast. Compare it with the previous day at the same times: if the pressure was also falling by half a hPa then, part of the fall is the daily cycle.
3. Write the forecast *before* seeing the next day, with a confidence from 1 to 5. A forecast written after the fact is not scored.
4. The next day, record the actual weather using the criterion fixed on day one, and tick whether the forecast was correct.
5. At the end, compute three percentages and compare them: the class's score, the lazy forecast's score ("tomorrow like today"), and the official forecast's score over the same days.
6. **Extension:** plot the whole two weeks of pressure as one line, with time on the horizontal axis, and mark the rain days on it. The link between the falling slopes and the marks is visible to

the naked eye and is the most convincing graph of the whole project.

6.3 Comparing two days

Open two dated exports side by side. Align them by *time of day*, not by row number: the two sessions can have different lengths. Plot both days' pressure on the same chart, with time of day on the horizontal axis, and do the same for temperature.

Three questions do almost all the analytical work. Did the whole curve shift up or down — so did the air mass change? Did the *shape* change, for example a steeper morning rise — so heating, or the number of people? And is the difference between the two days bigger than the wobble within a single day? Teach students to compute the daily mean and the daily range and to compare those two summary numbers before looking at any individual reading.

6.4 Questions to ask afterwards

1. Our pressure was 1008 hPa and it was clear outside, but a shop-bought wall barometer pointed to "storm". Who is wrong, and why? What number about our school would we need to know to answer?
2. The relative humidity indoors is 30% and outdoors it is 85%, but the dew points are nearly equal. What does that tell you about where the indoor air came from?
3. Our station reads 2 °C warmer than the official forecast, every day. Is the sensor broken? How would you find out?
4. The altitude card changed by 15 m overnight, even though the box was not touched. What happened, by how much did the pressure change, and what does that tell you about using a barometer as an altimeter?
5. You have to forecast tomorrow using only today's data from this station. Which single number would you choose, and what would you still be unable to predict with it?
6. Your forecast was correct on 7 days out of 10, and the lazy forecast "tomorrow like today" was correct on 6. What did you actually gain from the pressure tendency?

6.5 Ideas for assessment along the way

- **Low stakes:** the student receives **one day's pressure chart, already drawn**, marks the steepest section on it, and states, in words, what weather it predicts. No calculation and no number is required.
- **Standard:** a two-week weather log — the daily table, a combined pressure chart with the rain days marked, the forecast score, and a paragraph on how well the pressure tendency does *on its own*.
- **Extension:** a written comparison between the class station and the official forecast, with the bias expressed numerically and explained through instrument exposure and microclimate, plus a reasoned proposal for a better site. Strong groups **check the Analysis tab's pressure row**: they recompute the sea-level pressure and the three-hour tendency from the exported file, and show with numbers why the 1013 hPa threshold could not work at their school.

Common pitfalls and misconceptions

Using the 1013 hPa threshold as a forecast. The central mistake of this activity, encouraged by many wall barometers and by older versions of the app. 1013 hPa is only the standard sea-level value; what matters is the *direction* and the *speed* of the change, and the station value also depends on the school's altitude. The **Pressure** row of the **Analysis** tab

is discussed in class, not memorised.

Reading a tendency over ten minutes. Real tendencies are 0.1–0.3 hPa per hour; in ten minutes nothing shows, and what does show is noise or the daily cycle. The tendency is judged over three hours.

Direct sunlight on the box. A sensor in the sun measures its own warmed casing and can read several degrees high. Shade it, always, and never site the station on a sunlit sill.

Reading immediately after moving the box. Temperature and humidity need several minutes after a large change. A box carried in from outdoors and read at once still reports the outdoors.

Breathing on the sensor. Exhaled air is about 34 °C and nearly saturated. One breath sends humidity and dew point up sharply and takes minutes to clear. Breathing *near* the box, from 30 cm, is a demonstration; breathing into it during a reading is a ruined measurement.

Trusting the absolute altitude. It assumes today's sea-level pressure is 1013.25 hPa, which is almost never true. Height *differences* are trustworthy; the absolute value is not.

Confusing relative humidity with the amount of water. Air at 30% RH indoors and 85% RH outdoors, on a winter day, can hold almost the same amount of water. The dew point settles the discussion on the spot.

Moving the station halfway through the project. It destroys the comparability of the days. If it happens anyway, record the date and treat the series as two separate series — do not pretend it did not happen.

Comparing days without recording conditions. If nobody wrote down that the window was open on Thursday, Thursday's data can never be explained.

Relying on the automatic log. Files in the `logs` folder are no longer deleted automatically, but a colleague can switch deletion on in **Settings**, and a memory stick can be lost. What matters is exported on the day, into the shared folder.

Changing the "it rained" criterion halfway through the project. The score becomes meaningless. The criterion is fixed on day one and written on the board.

Safety and classroom management

Low overall risk, but the fixed station and the outdoor work each need a little thought.

Equipment left running unattended. Agree the position with the building manager, label the box with the class and your name, and never place it on an escape route, on a sill it could fall from, or where a door can swing into it. Anything left overnight sits on a socket the site staff know about.

Cables. Extension leads never cross a walkway. Tape them down or route them along a wall, and never daisy-chain extension leads.

Windows. The ventilation investigation needs open windows. Above the ground floor, only an adult opens windows, restrictors stay engaged, and nothing is placed on an open sill.

Outdoors and stairs. Electronics never go out in the rain. Supervise groups leaving the building. For investigation 3, the one about the building's height, the same rule applies as in the activity *Pressure and Altitude*: **nobody reads a screen while walking**; one student carries the box, another reads it, and only while standing still on a landing.

The daily routine. The duty pair does not enter a locked room alone and does not climb on a chair to reach the station. Choose a spot that can be reached standing on the floor, from the start.

Adaptations for pupils with special educational needs

The trend chart, already drawn. This is the support route and it reaches the same scientific conclusion. Do not ask the student to compute the tendency: **print one day's pressure chart, already drawn** (press **Pause** in the **Live Charts** tab and take a screen-

shot, or draw it once from the exported file and copy it), and ask only for the **interpretation in words**: rising, falling or steady, where the steepest part is, and what weather it predicts. The sheet has, for exactly this purpose, a row with the three arrows to tick and a frame for the received chart. The student reaches the same statement about the weather, by a different route.

Rows filled in in advance. Write the dates and times yourself in the daily table *before* copying the sheet, so the student only fills in the values; for extra support, also fill in the whole of the first day, including the room-state row — the student continues a pattern rather than inventing one.

Three real roles in the duty pair. The roles rotate weekly and none of them requires moving through the school:

- **routine keeper** — checks at the fixed time that the reading was taken, that the file was renamed correctly and that the room-state row is filled in; they have the right to declare a day "incomplete", so they decide what data enters the analysis, which is the role with the greatest scientific weight in the class;
- **screen reader** — says the five values out loud and dictates them;
- **weather chronicler** — records every morning how yesterday's weather actually was, using the criterion fixed on day one, without looking at the written forecast. This separation is what keeps the score honest.

Oral answers instead of writing. The daily forecast, the explanation of the dew point and the conclusion can be given **out loud**, with a classmate or the teacher writing them on the sheet. For students who find writing difficult, **ticking an arrow and circling a word are a fully accepted answer** in place of a sentence.

Language and cognitive support. Write on the board, on day one, the four words of the project — pressure, tendency, dew point, exposure — and use them unchanged for the two weeks. Repeat the same sentence every time: *what matters is which way it is going, not what it is.* For students who need predictable routine, the fixed reading time is an advantage, not a constraint: name them for it and leave them the same task every day.

If things do not go to plan

You have one box and thirty students. This is actually the activity's normal setup: there is one station, and the class rotates around it. Project the screen or seat the class in a semicircle, and at every reading a different student reads out the five values while everyone records them in their own table. For the comparison investigations, run them as a single class movement, with every group recording at the same stops.

You have twenty minutes, not fifty. This is exactly the short route. Keep exactly three moments: the prediction, the breathing from 30 cm with the jump in humidity and dew point, and the trip outside that shows the dew point does not move. Drop the export, the storey climb and the tendency.

You have no projector. A semicircle around the laptop, close enough to read the cards, and a **screen reader** named for each moment, who says out loud what the numbers are doing: "humidity is rising", "the dew point stayed put". The shape of a curve carries very well in words, and describing it out loud is exactly the skill being targeted.

The sensor will not settle. Ten minutes after any move is normal. If the values still drift after a quarter of an hour, the cause is almost certainly exposure: direct sun, a radiator, a laptop's vent or a draught. Move the station, record the date of the move, and restart the series.

The box will not connect on a project day. Do not stop the project. Record the day as missing — **a missing day recorded is data; a missing day hidden is a lie** — and carry on the next day. The gap shows on the chart and is worth discussing. Meanwhile,

check the cable, the port selected in the app and, if you are on WiFi, the network.
You cannot run the two-week project at all. Use the "What the data should look like" table in this guide, and a printed pressure series from the national weather service's website, as a ready-made data set: students compute the tendencies, write the forecasts, compare them with what happened and with the lazy forecast. All the reasoning stays the same; only the students' own measurement is lost.

7. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are root addresses, not deep links, so they will not rot over time.

- **National Meteorological Administration (ANM)** — `https://www.meteoromania.ro` — the daily forecast to score the class's forecasts against, the surface pressure chart and county station data; the external reference for the whole project.
- **NOAA** — `https://www.noaa.gov` — search for pressure systems, fronts, ground-observation standards and the difference between station pressure and pressure reduced to sea level; explains exactly why the 1013 hPa threshold does not work at altitude.
- **Copernicus** — `https://www.copernicus.eu` — European reanalysis and climate data, to place two weeks of school readings in a longer context.
- **NASA Global Climate Change** — `https://climate.nasa.gov` — the distinction between weather and climate, the natural closing discussion of the project.
- **PhET Interactive Simulations** — `https://phet.colorado.edu` — search for gas properties and states of matter; support the ideas of pressure and saturation before the measurement.
- **Institute of Physics** — `https://www.iop.org` — search for pressure, gas laws and measurement uncertainty; teaching materials suited to the upper-secondary end of the activity.

8. Assessment

The exit ticket, identical to the one on the student sheet: *one thing that surprised you today, and one thing you would do differently if you repeated the measurement.* It is collected in the last two minutes of the lesson and says more about the class's understanding than any mark.

For a more formal judgement, use the **shared three-strand rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. Because the activity spans two levels, the descriptors read as follows:

- **Measurement practice** — the student reads at the fixed time, records the room's state alongside the number, exports and names the file correctly, and does not move the station. At Scientist level, in addition: checks the instrument's exposure before questioning the reading.
- **Data interpretation** — at Explorer, the student correctly states whether the pressure is rising, falling or steady and what weather that predicts. At Scientist, they compute the tendency in hPa per hour over three hours, compare it with the operational thresholds and with the daily cycle, and score their forecasts against the lazy forecast.
- **Scientific reasoning** — the student explains why the *tendency*, not the 1013 hPa threshold, is what matters, and uses the school's altitude to show why; distinguishes relative humidity from the amount of water using the dew point; and distinguishes "our method added nothing" from "pressure has nothing to do with weather".

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

9. The student sheet

The sheet the students fill in is a separate document, in the folder `docs/en/student_sheets`: the file `weather_station.pdf`. It has two pages — the prediction, the chart frame and the daily recording sheet on the first; the tendency calculation, the forecast log and the conclusion on the second — so it prints double-sided, one sheet of paper. It is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write — and asks for the prediction **before** the first day of measurement.

Four things worth knowing before printing it. **The sheet carries the rows for a single week of duty**, not the whole project: the complete series lives on the wall chart and in the cumulative file, and every duty pair gets a fresh sheet. **The forecast log has a column for the tendency in hPa per hour and none for the value compared with** 1013: the shape of the sheet says the same as the **Analysis** tab — it is the trend that matters, not the threshold. **The frame for the received chart** sits on the first page, ahead of any calculation, with the three arrows to tick — rising, falling, steady — precisely so that the support route starts from interpretation, not from arithmetic; that is where you glue in the chart you printed. **The room's state has its own column right in the reading row**, not a box at the end, because a condition recorded an hour later is a condition invented. The tables are not repeated here precisely so that the two cannot drift apart: there is a single printable version, and it is the sheet.